Hugo Marquerie 04/03/2025

# Exponente complejo

**Definición 1 (Exponente complejo).** Sean $\alpha, \beta \in \mathbb{C}$ con $\alpha \neq 0$, $\alpha^\beta$ es $\alpha$ elevado a $\beta$

$$\iff \alpha^\beta = \left\{ w \in \mathbb{C} : w = e^{\beta \log \alpha} = e^{\beta(\ln|\alpha| + i(\operatorname{Arg}(\alpha) + 2k\pi))} : k \in \mathbb{Z} \right\}.$$

Por abuso de notación, escribimos $\alpha^\beta = e^{\beta \log \alpha} = |\alpha|^\beta \, e^{i\beta(\operatorname{Arg}(\alpha) + 2\pi k)}$.

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